Smart Real-Time Age and Emotion-Based Advertising in Malls Using Custom-Built Models

Project ID: 25-26J-091

Project Proposal Report

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Department of Information Technology

Bachelor of Science Special Hons Degree in Data Science

Sri Lanka Institute of Information Technology

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A Demographic Driven Advertisement Recommendation Engine Incorporating Serendipity and Demographic Filtering

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August 30

DECLARATION

I declare that this is my own work & this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning & to the best of my knowledge & belief it does not contain any previously published or written by another person except where the acknowledgement is made in the text.

Name	Student ID	Signature
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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

Nathali-A

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Signature of the supervisor

25/08/2025
Date

Ms. Shashini Kumarasinghe
Signature of the co-supervisor

Date

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ACKNOLEDGEMENT

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Finally, I am greatly indebted to my parents for their unconditional love, support, and encouragement, which have been a source of inspiration during this time.

ABSTRACT

This research focuses on the challenges of static and fixed-schedule digital ads in Sri Lankan shopping malls. It aims to create an AI-powered advertisement recommendation system that responds to real-time audience demographics and emotional states. Instead of using pre-recorded CCTV footage, the system relies on live video streaming as its main data source.

It employs computer vision techniques to identify non-identifiable audience traits like estimated age group, gender, and mood, which could be happy, neutral, or sad. The system processes these inputs using collaborative, content-based, hybrid, and demographic filtering methods. The recommendation engine picks ads from a categorized database and includes serendipity-focused and diversity-aware filtering.

This approach avoids ad fatigue while staying relevant. Additionally, it incorporates a context-aware layer to adjust recommendations based on time, location, crowd composition, and engagement history. To tackle the cold start problem, the system uses group-level behavior and ad metadata to offer relevant suggestions for first-time users. By matching ad content with real-time audience profiles and the surrounding context through ongoing live stream analysis, this research aims to improve user engagement, brand recall, and advertising effectiveness. It also promotes privacy-conscious AI use in retail spaces. The final evaluation will look at how accurately the system detects the audience, the relevance and diversity of recommendations, and its overall effect on engagement levels in mall settings.

Keywords: Context-Aware Advertising, Real-Time Recommendation, Serendipity, Diversity-Aware Filtering, Cold Start Problem

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1. INTRODUCTION

The research provides a system for real-time ad recommendation based on artificial intelligence that exceeds traditional static displays by combining different intelligent elements with a cohesive framework. Key to this system are collaborative, content-based, hybrid, and demographic filtering methods. The combination ensures that ads reflect individual user tastes while being flexible to group-level audience attributes. Unlike ordinary methods, the system takes real-time data from live video broadcasts. Age detection and emotion recognition models and behavioral pattern tracking models dynamically observe the present audience profile at all times. The resultant information guides the recommendation engine to allow for dynamic delivery of contents that can quickly respond to shifts in audiences' moods and demographic attributes.

The major contribution of the system being proposed is its use of surprise when it comes to selecting ads. Presenting intentionally surprising yet pleasurable ads at all points boosts consumer engagement and creates points of surprise to build positive feelings towards brands. This element of surprise is closely tied to a diversity-sensitive filtering system. The filtering system assists the system to avoid repetition of ads while maintaining novelty, diversity, and balance in deliveries. This diversity itself reduces ad fatigue and is helpful to introduce new products to audiences who are neither seeking nor necessarily active towards them.

In order to augment these suggestions further, a context-aware layer is added to the framework. This is based on considering elements such as time of day, whereabouts in the mall, number of people to be targeted, and historic levels of engagement. This permits dynamically changing ads based upon circumstances. For example, evening priority can be given to family-based ads when there are more groups present, yet youth-based offers might be encouraged during weekends.

Another significant contribution relates to addressing the cold start problem through this system. The cold start problem arises in recommender systems when there is insufficient historical data to make relevant recommendations. For this work, cold start situations are avoided by using group-level behavioral analysis and filtering methods based on metadata. This approach enables the system to make relevant ad suggestions to first-time users as well as never-before-experienced audiences. The system achieves relevance by using high-level audience trends and contextual metadata. This way, it always achieves relevance at first glance.

Each one of these combined components—demographic and emotion-based input at run time, multi-technique recommendation, surprise with diversification consideration, context adaption, and cold starts solutions—amount to a complete solution optimized for retail mall environments. This maximizes ad appeal and memorability and provides tangible value to advertisers. They get a more convenient and consumer-centric way to promote brands.

1.1. Background

The accelerating advancement of computer vision (CV) and artificial intelligence (AI) has increasingly changed ad delivery in digital and physical spaces. The customary ad methods in shopping malls mostly depend upon static posters, looped digital signage, or non□adaptive campaigns that are neither responsive to altering viewer demography nor situation contexts during their presence. The latest findings from demographic recognition and affect analysis now make it possible for ad systems to determine character and affective states of observers close to a display and present individually adapted and situationally appropriate recommendations accordingly. Studies have evidenced such responsive systems to enhance ad recognition, user interaction, and consumer satisfaction when compared to customary methods [1].

Parallel research on machine learning (ML) and recommendation has overcome boundaries from e-commerce and spread to physical spaces such as retailing malls. Though collaborative filtering and content-based recommendation methods are still mainstream approaches, they are faced with challenges when it comes to cold-start situations and group environments. Here, demographic filtering stands out as a strong alternative by offering personalization with minimum prior user information requirement [2]. Moreover, incorporating serendipity—balancing relevance and novelty—also helps to reduce ad fatigue and thus sustain audience interaction with promotional materials [3].

Based on these advances, work is underway to integrate with IoT and digital signage networks, whereby edge or cloud servers process in real time demographic and emotion information to provide personalized ads dynamically. Yet privacy protection, fairness, and cultural flexibility are concerns to date, and solutions are needed that are at once technically sound and ethically informed [4]. The present system makes its contribution to this literature by bringing together demographic profiling and emotion recognition with reinforcement learning and serendipity-driven personalization, with a specification to mall environments.

1.2.Literature Survey

1.2.1. Demographic and Emotion Recognition in Advertising

Computer vision-driven demographic recognition systems predict attributes like age, sex, and group makeup with deep learning networks like CNNs and ResNet. They are effective at high accuracy with changing illumination and viewpoints from any camera and are ideal for crowded locations [5]. Emotion detection takes personalization to another level through facial expression analysis (e.g., happy, sad, neutral) with facial action coding systems or deep CNNs [6]. Research indicates that demographic and emotion intelligence permit real-time customization like offering youth-centric products to youngsters or offering family bundles when families are identified.

1.2.2. From Static Campaigns to Context-Aware Advertising

Previously marketed to with fixed schedules and mass-based targeting, there was little relevance and minimal engagement. Context-aware ad systems, by contrast, dynamically adjust campaigns to actual-time audiences' attributes and reactions [7]. Studies confirm contextual targeting dramatically enhances ad effectiveness, recall, and buying intent over static campaigns [8]. Further, through emotion feedback, it is possible to have systems gauge reactions (e.g., smiling, interested stare) and adjust subsequent recommendations at run time.

1.2.3. Machine Learning for Real-Time Ad Recommendations

Machine learning is the core element of adaptive ads, using methods like classification models, collaborative filtering, and reinforcement learning. More specifically, reinforcement learning provides for system improvement over time through absorbing information from past audience reactions to ads [9]. Predictive models determine the chances of interaction with certain ad types and thus help advertisers maximize their efficiencies over time. By combining demographic and affect data with adaptational algorithms, these systems find a trade-off among accuracy, novelty, and individualization.

1.2.4. Integration with IoT and Digital Signage Systems

Contemporary mall promotion blends with IoT-powered digital signage networks, whereby kiosks or ceiling mounts embedded with sensors and cameras collect audience data. Processing is done locally on edge devices (e.g., Raspberry Pi, Jetson) or cloud servers with minimal latency and privacy maintenance [10]. This edge-to-edge feedback loop—capture, process, recommend, and show—allows ongoing ad optimization at a time interval of seconds.

1.2.5. Limitations and Ethical Considerations

Despite these benefits, there are certain challenges to be faced. There are privacy and biometric data-specific issues with the need for anonymization and protocol adherence like GDPR [11]. Algorithmic bias when there is demographic recognition can lead to biased ad-targeting or insufficient cultural sensibility [12]. Lighting change, occlusion by masks and spectacles, and cultural expression all affect performance and require ongoing retraining of models. Transparency and fairness and ethical deployment are therefore key to public acceptability.

1.2.6. Proposed System Context

The system outlined herein combines these innovations through the application of demographic profiling, emotion recognition, and reinforcement learning-guided ad recommendation to an in-mall digital signage network at a real-time scale. Differing from schedule-based advertising, it will select promotional materials most relevant to audiences actively being surveyed and adjust subsequent recommendations based on actual-time feedback for engagement. It is intended to improve ad relevance, customer experience, and advertiser return-on-investment through data-driven personalization.

1.3.Research Gap

Feature / Objective	Existing Systems (Research)	Proposed Application Component
Personalization Depth	Most mall ad systems rely only on basic demographic filtering (e.g., age or gender), leading to shallow targeting.	Combine multiple demographic factors (age, gender, mood, group context) for rich personalization.
Serendipity	Current ad engines focus on pure relevance, causing predictable and repetitive ads → ad fatigue.	Introduce serendipity (relevant but novel ads) to keep content fresh, surprising, and engaging.
Cold-Start Problem	Collaborative filtering and history-based systems fail without prior user data.	Apply demographic-based filtering to handle cold-start scenarios effectively.
Engagement Measurement	Existing systems often do not evaluate real-time engagement (e.g., dwell time, ad recall).	Use behavioural metrics (dwell time, attention, recall) to dynamically improve ad selection.
Real-Time Adaptation	Ads are often pre-scheduled or updated slowly, not responding to immediate audience changes.	Deploy a real-time recommendation engineadapting ad content instantly to changing demographics.
Privacy Handling	Some research prototypes store raw video or identifiable data, raising compliance issues.	Extract only non-identifiable features (age group, gender, mood), anonymizing all data in line with GDPR.

Integration with Security	Current systems are advertising-only, missing links with security or behavioural anomaly detection.	Provide a dual-purpose system with both advertising personalization and potential for suspicion detection.
User Experience	Passive, one-way advertising can feel intrusive or boring.	Deliver interactive, adaptive, and personalized ads, making the experience engaging and relevant.

Today's mall ad platforms are either weak demographic filters or software-based digital signage systems with rule-driven ad changing that fail to deliver depth, uniqueness, and interactivity. They offer repetitive messaging, serendipity is lacking, and there is no ability to move audiences dynamically in real time. To all this is added poor management of privacy and ethics by many installations.

The present work bridges this gap by proposing a real-time, demography-driven ad recommendation system using age, gender, mood, and group context to make it personalized and unique. The system uniting serendipity and privacy-respecting feature extraction along with real-time measurement of engagement makes ads relevant and surprising yet never tiresome and consistently trustworthy to its user base.

1.4.Research Problem

In today's ad environment, our ability to present targeted and relevant information to varied sets of users is not simply desirable but ever more critical. Marketers desire to maximize return on investment while guaranteeing that consumers find ads valuable and less interruptive. However, traditional ad methods rely heavily on fixed profiles or past history, such as buying habits, browsing activities, or broad-based demographic segmentation. However useful these methods have been to create comprehensive recommendation programs inside net platforms, their application to brick and mortar retail environments—i.e., shopping centers—remains significantly neglected. As a consequence, there is a gap between the copiousness of behavioral signals available in physical environments and relatively rigid, one-dimensional ad strategies being utilized at present.

One critical drawback with state-of-the-art systems is their static filtering approach. Commercial recommendation sites mostly adopt collaborative and content-based filtering, yet these are essentially backward-looking methods. They rely upon past user interactions or item similarities and thus react with a lag to fresh user mood swings, situations, or tastes in actual environments. This lack of adaptability is partly responsible for a widely known issue commonly termed ad fatigue, whereby users are subjected to ever-predictable or all-too-personal ads that lose potency with time. Apart from this, static systems rarely consider emotional or group-level phenomena, which are crucial in group environments like shopping plazas, where decisions are commonly family-wide, with their groupmates/family/friends, or by crowd consensus.

A critical gap can be observed in aligning personalization and serendipity. Personalization secures relevance by tailoring ads based on observed or predicted individual preference; however, alone it runs the risk of creating an "echo chamber effect." Long term, users are consistently subjected to the same type of ad with little surprise factor, breaking user interaction and engendering a lack of curiosity. Research studies performed on recommendation tools point out that users are pleased not only with relevance but with diversity and surprise—factors evoking curiosity, driving discovery/exploration, and increasing ad message retention. Unfortunately, most present demographic-/content-based recommendation tools prioritize accuracy to the detriment of diversity and hence long-term satisfaction suffer accordingly. The opportunity to pursue a serendipity-based approach—users receiving ads relevant and yet pleasingly surprising—remains comparatively untapped when considering physical retail adverting space.

A third research gap is with respect to using insufficiently comprehensive sets of recommendations based upon dynamically obtained real-time data regarding population demographics, emotions, and behaviors. Newly emerging fields like affective computing and computer vision provide techniques for deriving non-identifiable attributes like age, gender, mood state, and group environment while preserving privacy. However, despite these innovations at a technical level, most ad systems continue to overlook dynamic signals potentially capable of greatly enhancing targeting effectiveness. For example, a customer visibly interested or excited by an environment may respond to displays differently from someone who looks neutral or bored. Similarly, behaviors observed by different groupings—e.g., families with kids versus groupings by young adults—offer contextual information potentially helpful for selecting apposite ads to show to them. These dynamic attributes are yet to find mainstream incorporation within retail-predominant ad-engines, however, and thus disclose an imbalance by highlighting what is possible at a technical level compared to implementation at a commercial level.

Additionally, prior work highlights continued challenges with cold-start cases—scenarios where there is little or no antecedent data to draw upon for an individual user. Most modern systems either poorly serve new users or fall back to using non-personalized, generic ads. In high-traffic locations like shopping malls, where many visitors are first-time buyers, this limitation generates significant lost interaction opportunity. Though collaborative and hybrid schemes have acted as targets for research to mitigate the cold-start problem in online communities, these ideas have yet to mature fully as applied to real-time group-level demographic filtering. This leaves open an opportunity to create systems to take advantage of raw crowd-level data—age dispersion, group size, or shared behavioral propensity—to make educated, privacy-conscious ad suggestions regardless of personal histories.

Thirdly, there is a user experience gap. Classic in-store ad mechanisms are usually deemed intrusive or irrelevant when users feel "watched" with no value add. Passive monitoring systems with no significant interaction are likely to lose audiences. There is increasingly evidence to show, however, that engagement-driven methods—such as interactive screens or gamification—can flip the ad perception from being observed to being entertainable. Very little research has investigated the possibilities of combining privacy-aware demographic analysis with interactive serendipity-driven ad recommendation to balance user trust with advertisers' objectives.

To overcome these lacunae, we propose a context-aware, hybrid recommendation system combining collaborative filtering, content-based filtering, demographic information, and serendipity-driven techniques. The system adopts real-time signals involving demographics, emotions, and behavior and at the same time protects user privacy while seeking to change ads dynamically based on external factors like time, space, audience size, and engagement levels. The system combines a diversity-conscious approach and reconciles personalized demand with diversity demand to ensure ads are relevant and at the same time surprise and entertain users while reducing chances of fatigue. The system further overcomes cold-start challenges by utilizing group-level analyses and combining metadata with contents and thus ensuring novice visitors are exposed to relevant and interesting ad presentations.

In essence, this research seeks to provide a wiser, more attuned ad experience—one that maximizes relevance, enhances freshness, and drives long-term engagement. By combining predictive analytics, serendipity logic, and affective intelligence, the proposed system addresses fundamental shortcomings of current advertising models, thereby offering a pathway toward a new generation of real-time, privacy-conscious, emotionally aware recommendation engines for physical retail environments.

2. OBJECTIVES

2.1. Main Objectives

To design and evaluate an intelligent, personalized advertisement recommendation system that integrates collaborative, content-based, hybrid, and demographic filtering with real-time age, emotion, and behavior detection to deliver context-aware, diversified, and serendipitous ad experiences to maximize user engagement and advertiser value.

2.1. Specific Objectives

- **Leverage Demographic Filtering:**

Integrate real-time detection outputs—age, gender, and group-level behaviors—into the recommendation framework to greatly enhance precision and relevance for targeting. Fueled by using demographic information as its base, it makes ads relevant to those audiences present and thus increases levels of engagement and campaigns' effectiveness.

- **Implement Serendipity and Diversity-Aware Logic:**

Introduce mechanisms that occasionally present unexpected yet relevant advertisements alongside personalized recommendations. This prevents user fatigue from repetitive ads, sustains novelty, and encourages curiosity, ultimately extending user interaction and fostering long-term engagement.

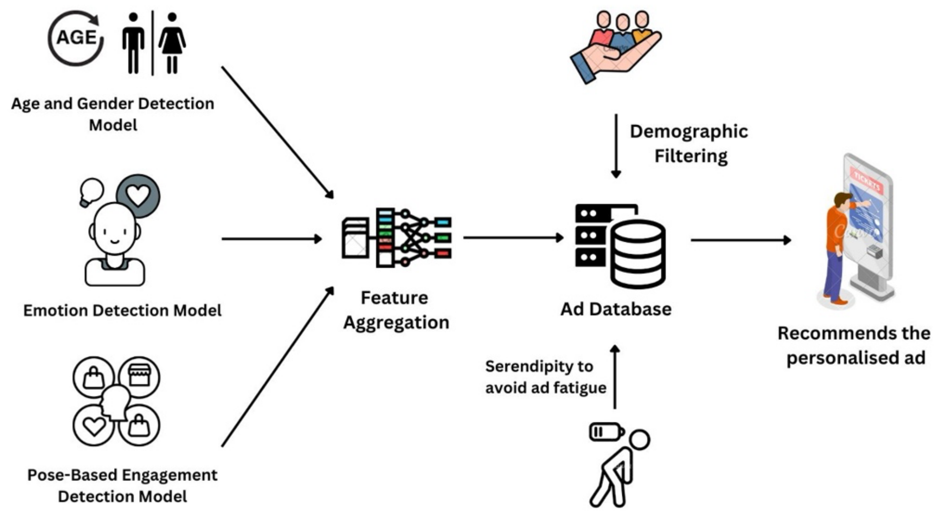
- **Enable Context-Aware Adaptation:**

Tweak recommendations continuously according to environmental and contextual variables like time of day, location at the mall, group size, and past audience reaction. This makes ads relevant at changing circumstances and enhances resonance with audiences and chances for favorable reaction.

- **Mitigate the Cold Start Problem:**

Estimate useful suggestions to occasional or new users from collective behavioral data, population data, and ad metadata. This approach makes it possible to always have the system supply valuable and attractive information irrespective of individual user history, hence overcoming one of the main challenges to recommender systems.

3. SYSTEM METHODOLOGY



Derive relevant recommendations for first-time users or infrequent users by utilizing group-level behavioral data, crowd demographics, and ad metadata. This makes it possible for the system to still present valuable and relevant content when there is no user history to tap into, overcoming one of the main challenges of recommender systems.The approach to this research step is to combine human–computer interaction principles with computer learning technology to facilitate a privacy-conscious, interactive system allowing for real-time age estimation and context-aware promotional displays at shopping malls. Unlike traditional passive static facial detection techniques only involving photography, this approach uses an interactive game-based interface to promote active user participation with provision of superior quality facial information and better detection. The approach lays out step-by-step process from conceptual design to deployment with iterative examination and data-driven improvement and ethics. It makes provision for using computer vision models streamlined towards edge deployment, application of privacy-conscious measures like facial embeddings, and construction of a dynamic ad recommendation pipeline. By means of systematized phases of requirements examination, system design, construction of models, validation, and field examination, this approach ensures the system to be sturdy, scaleable and robust with divergent environmental conditions while meeting the goals of accuracy, reaction time, and user acceptability.

3.1. Application of Key Pillars within the Specialized Knowledge Domain

This study draws on four specialized domains to achieve its objectives:

- **Machine Learning and Recommendation Systems:**
  Collaborative filtering, content-based filtering, and hybrid methods will be employed to generate personalized recommendations. Demographic filtering based on real-time signals will complement these approaches, while serendipity and diversity mechanisms will ensure novelty and prevent ad fatigue.
- **Real-Time Computer Vision:**
  Computer vision and deep learning models (e.g., CNNs) will be used for age, gender, and emotion detection, with tools like OpenCV and MediaPipe enabling efficient deployment. Outputs from these models will serve as dynamic inputs to the recommendation engine.
- **Context-Aware Personalization:**
  Recommendation logic will incorporate situational factors such as time of day, crowd size, location within the mall, and historical engagement data, ensuring that advertising aligns with the immediate context of the audience.
- **Privacy-Preserving AI:**

To ensure compliance with GDPR and ethical standards, the system will process only non-identifiable features such as embeddings, avoiding the storage of raw images. Edge-based computation (e.g., Jetson, Raspberry Pi) further minimizes data transmission risks.

3.2. Application of Technologies in the Field of Application

The system will integrate modern frameworks and hardware for robust, real-time operation:

- **Recommendation Frameworks:** LightFM, Surprise, or custom-built hybrid models for collaborative and content-based filtering.
- **Computer Vision Tools:** OpenCV, TensorFlow, or PyTorch for training and deploying demographic and emotion detection models.
- **Real-Time Detection:** Pretrained models such as RetinaFace, MTCNN, or MediaPipe for facial localization and feature extraction.
- **Backend & APIs:** Python (Flask/FastAPI) or Node.js for serving recommendations, linked with scalable databases (PostgreSQL/MongoDB).
- **Deployment Hardware:** Edge devices such as NVIDIA Jetson or Raspberry Pi for low-latency processing and privacy-preserving operation.

These technologies are selected to balance computational efficiency, scalability, and cost-effectiveness while supporting practical retail deployment.

3.3.How to Carry Out the Project

The project will follow an iterative development lifecycle:

1. **Requirement Analysis & System Design:** Define system specifications, user needs, and ethical/privacy constraints.

2. **Prototype Development:** Build a minimal version of the detection and recommendation modules.
3. **Model Training & Optimization:** Train hybrid recommendation algorithms and fine-tune vision models for local mall conditions.
4. **Integration of Context-Awareness & Serendipity:** Implement novelty-driven logic to balance personalization with diversity.
5. **System Integration:** Connect detection, recommendation, and interface modules into a complete workflow.
6. **Testing & Validation:** Conduct controlled tests and pilot mall trials to measure accuracy, latency, and engagement.
7. **Evaluation & Refinement:** Compare system performance against baseline recommendation approaches and refine based on user feedback.
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3.4.Tasks and Sub-Tasks Identified to Meet Objectives

- **Demographic & Emotion Detection Module**
  - Train and validate age, gender, and emotion models.
  - Optimize for real-time inference on edge devices.
- **Recommendation Engine**
  - Implement collaborative, content-based, and hybrid filters.
  - Integrate demographic and context-aware inputs.
  - Add serendipity and diversity-aware filtering.
- **Context-Aware Layer**
  - Implement time, location, and group-level behavior adaptation.
  - Ensure fallback recommendations when context signals are unavailable.

- **System Integration & Backend Services**
    - Develop API services for seamless data flow.
    - Link detection and recommendation modules with signage displays.
- **Privacy & Security Measures**
    - Implement embedding-based processing.
    - Ensure compliance with data protection regulations.
- **Testing & Evaluation**
    - Conduct lab tests for accuracy and latency.
    - Run field trials in mall-like environments.
    - Collect engagement and user satisfaction metrics.

3.5.Data Requirements and Collection Methods

The project will require heterogeneous datasets combining facial images, demographic annotations, and emotional expressions:

- **Training Datasets:** Publicly available datasets such as UTKFace (age/gender), AffectNet (emotion), and IMDB-WIKI (age).
- **Evaluation Data:** Real-world mall video feeds under varying conditions (lighting, occlusion, crowd density).
- **Feature Processing:** Data will undergo preprocessing including alignment, augmentation, and anonymization through embeddings.

All test data captured will be anonymized, with no raw images stored, ensuring compliance with privacy requirements.

3.6.Project Requirements

3.6.1. Functional Requirements

- The system will process and combine user interaction data specified by clicks, views, interaction time, and ad preference history.
- The system would make use of collaborative filtering, content-driven filtering, hybrid filtering methods to create personalized recommendations for ads.
- The reasoning for recommendations has to account for real-time feedback according to models to determine age, gender, emotion, and group activity.
- • A context-aware layer integrating recommendations with time, space, crowd makeup, and historical trends of engagements needs to be there for adjusting recommendations accordingly.
- Diversity-conscious and serendipity-oriented filtering mechanisms ought to periodically present innovative but pertinent advertisements to mitigate ad fatigue and maintain engagement.
- The system needs to solve the cold start issue based on group-level behavioral information and ad metadata to generate relevant recommendations to new users of the system.

3.6.2. User Requirements

- Users expect highly relevant and engaging ad recommendations tailored to their preferences, demographic profile, and real-time context.
- The system should occasionally surprise users with unexpected but enjoyable ads, enhancing engagement and ad recall.
- Users expect the interface to be visually appealing, simple to use, and responsive across different devices.
- The system should prevent ad repetition and fatigue by ensuring diversity while maintaining relevance.
- Even first-time users should receive accurate and contextually appropriate recommendations through cold start mitigation strategies.

3.6.3. System Requirements

- Compatibility with standard CCTV/IP camera inputs.
- Real-time recommendation generation with minimal latency.
- Scalable across multiple devices and locations.
- Robust operation under diverse environmental conditions.

3.6.4. Non-Functional Requirements

- All user data must be securely stored and transmitted using encryption, complying with relevant data protection regulations (e.g., GDPR).
- The system should be designed for scalability to handle growing user bases and large volumes of ad and interaction data without performance degradation.
- The recommendation engine should deliver results in real-time or with minimal latency to maintain a smooth user experience.
- The user interface should be intuitive, responsive, and optimized for various devices, including kiosks, digital signage, and mobile platforms.
- The system should provide transparent explanations or visual cues for recommendations to enhance trust and user acceptanc

3.6.5. Test Cases (tentative)

Test Area	Test Case ID	Description	Input Data	Expected Output
1. System Initialization & Interface	TC-1.1	Verify system starts and initializes correctly	System power on	Dashboard and modules load without errors

Test Area	**Test Case ID**	**Description**	**Input Data**	**Expected Output**
	TC-1.2	Check signage connection	Digital signage connected	Ads display successfully after handshake
	TC-1.3	Validate user interface simplicity	User interacts with admin panel	Intuitive navigation and correct module access
2. Demographic & Emotion Detection	TC-2.1	Verify real-time age detection	Face of age ~30	System predicts correct adult range
	TC-2.2	Validate gender detection accuracy	Male/Female faces	Gender classified correctly
	TC-2.3	Test emotion detection (happy, sad, neutral)	User expressions	Correct emotion label identified
	TC-2.4	Handle occlusion (mask, side profile)	Partially hidden faces	Reasonable demographic estimation with fallback
3. Recommendation Engine	TC-3.1	Validate ad recommendation based on demographics	Input: young adult male	Ad targeted for that demographic displayed
	TC-3.2	Test serendipity filter	Same demographic repeatedly	Occasionally novel but relevant ads displayed
	TC-3.3	Check diversity-aware filtering	Continuous detection of same group	Ads rotated to avoid fatigue
	TC-3.4	Cold-start handling	First-time user	Group-level or metadata-

Test Area	Test Case ID	Description	Input Data	Expected Output
			group with no history	based ads shown
4. Context-Aware Adaptation	TC-4.1	Validate time-based adaptation	Input: morning vs. evening	Ads change according to time of day
	TC-4.2	Test location-specific ads	Zone A (food court) vs. Zone B (fashion)	Relevant ads displayed per location
	TC-4.3	Validate group behavior detection	Group of families detected	Family-oriented ads shown
5. Data Handling & Privacy	TC-5.1	Verify raw image deletion	Video captured during detection	No raw images stored after processing
	TC-5.2	Ensure embedding-based storage	Face processed	Only embeddings stored in database
	TC-5.3	Test secure data transfer	Detection output to recommendation module	Data encrypted during transmission
	TC-5.4	Validate consent handling	User denies camera use	System halts ad personalization, fallback ads displayed
6. Performance & Reliability	TC-6.1	Check latency for ad updates	New demographic input	Ad updates within <1s
	TC-6.2	Validate system under load	Multiple users in frame	Accurate recommendations

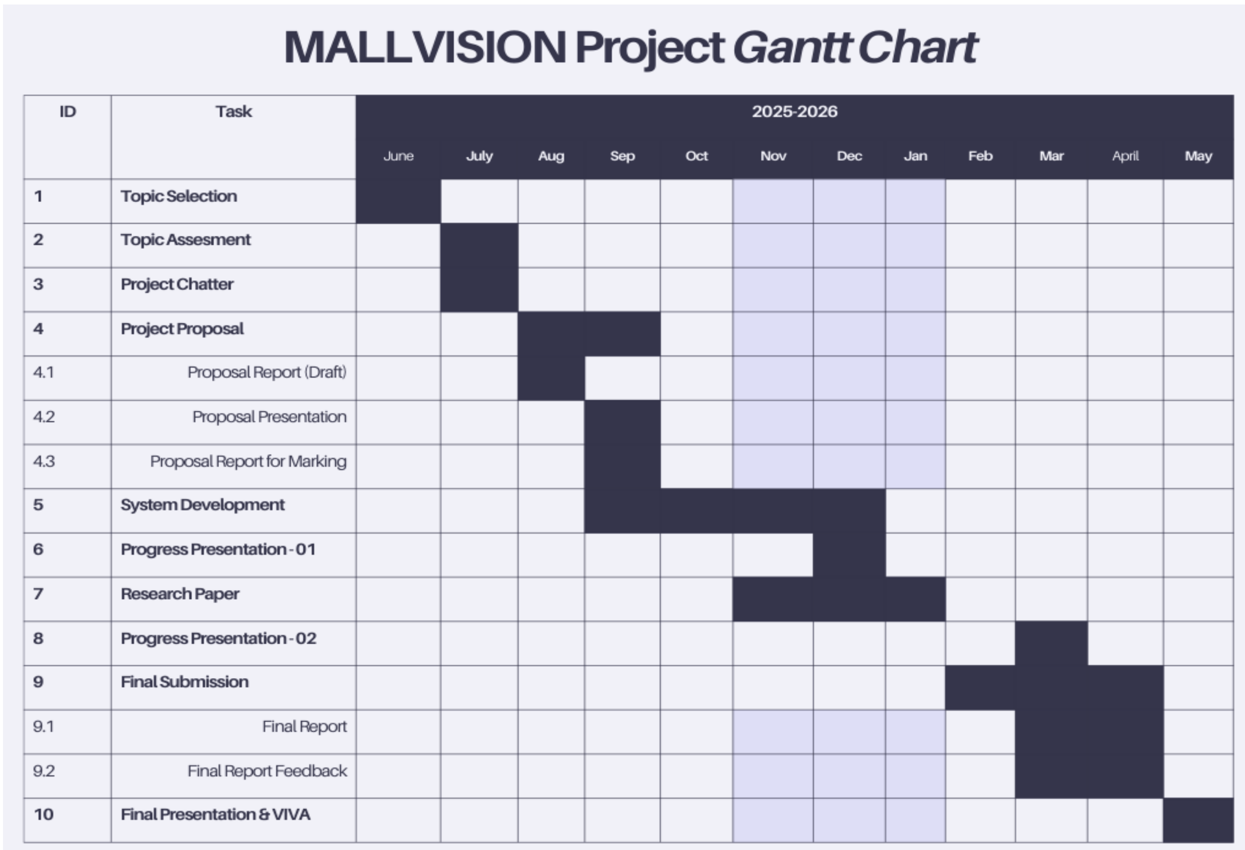
Test Area	Test Case ID	Description	Input Data	Expected Output
				without lag
	TC-6.3	Test fault tolerance	Temporary camera disconnection	System switches to default ad playlist
	TC-6.4	Uptime and recovery test	System restart during operation	Recovers and resumes without data loss

3.7.Anticipated Conclusion

By following this methodology, the project aims to deliver a context-sensitive, serendipity-enhanced advertisement recommendation system that maximizes engagement while safeguarding privacy. The system is expected to demonstrate:

- Improved personalization by integrating demographic, emotional, and behavioral signals.
- Sustained engagement through diversity and serendipity-based filtering.
- Real-time, scalable performance across shopping mall environments.
- Compliance with data protection standards, ensuring trust and ethical deployment

3.8.Gantt Chart



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4. DESCRIPTION OF PERSONAL & FACILITIES

The project contributions are a holistic approach to real-time advertising and security enhancement through combined age, emotion, and behavior recognition systems. Fonseka W. S. M. focuses on building the interactive game-based age recognition subsystem, dividing users accurately into age categories to help deliver targeted advertising. Samaraweera W D U I oversees the emotion detection module, detecting risky emotional indicators such as fear, stress, or aggression and keeping a list of potentially risky individuals. Abeykoon S N oversees the behavior detection capability, monitoring individuals' actions to validate the suspicions raised and sending safety alerts upon verification of high-risk behavior. Rajana H T R trains and deploys the recommendation engine based on join output from age, emotion, and behavior inference to deliver accurate, privacy-protecting advertising to various audiences in real time.

4.1. Facilities

The project utilizes the computational resources at the university, including high-performance workstations with GPU capability for model testing and training. Commercially available surveillance cameras and edge light devices simulate a normal real-time shopping mall environment. Software tools utilized are Python-based machine learning libraries (TensorFlow, PyTorch), real-time video processing toolkits (OpenCV, MediaPipe), and cloud platforms for model serving and testing.

This collaborative setup ensures seamless integration of all components, providing a robust and scalable system that can enhance both personalized advertising and mall security management simultaneously.

5. COMMERCIALIZATION ASPECTS OF THE PRODUCT

The proposed **Demography-Based Advertisement Recommendation Engine with Serendipity** is designed to enhance customer engagement and improve advertiser ROI by showing contextually relevant yet non-repetitive ads. By combining demographic filtering, real-time analytics, and novelty-driven recommendations, it stands out as an innovative, user-friendly, and business-focused solution.

Market Space

1. Advertising Market
   Digital-Out-of-Home (DOOH) advertising is growing rapidly, driven by demand for measurable engagement and targeted campaigns. The proposed system aligns with global trends toward personalization, diversity-aware content, and responsible AI use.
2. Market Size

The targeted advertising and retail analytics sector is valued in billions globally, with Asia-Pacific (especially malls and urban retail) being a prime growth market. In Sri Lanka and similar economies, AI-based targeting is emerging, creating strong opportunities for low-cost, scalable, and privacy-conscious solutions. Existing Systems

3. Existing Systems

Current systems rely heavily on static digital loops or basic demographic targeting, leading to repetitive ads and ad fatigue. They also lack novelty, contextual adaptation, and effective cold-start solutions. The proposed system differentiates itself by integrating serendipity to balance personalization with freshness.

4. Competition

Competitors include DOOH solution providers and AI-driven ad platforms. However, very few combine demographic filtering with serendipity and context-awareness. Our system offers better engagement and cost-effectiveness through diversity-aware personalization and edge-device scalability.Revenue Model and Commercial Strategy

5.2. Revenue Model and Commercial Strategy

- Licensing Model – Malls, ad agencies, or retail chains purchase licenses to deploy the system across multiple locations.
- Subscription Model (SaaS) – Affordable subscriptions for small retailers, offering real-time dashboards, analytics, and premium engagement metrics.
- Advertising Partnerships – Direct brand collaboration, charging per impression, engagement, or dwell-time, supported by serendipity-driven targeting.
- Hardware + Software Bundling – Low-cost kiosks and signage units bundled with software for markets lacking infrastructure.
- Anonymized Analytics as a Service – Aggregated, non-identifiable demographic and engagement insights sold to advertisers for campaign optimization.

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